

# LAKSHMIDHAR KOTIPALLI

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## PROFESSIONAL SUMMARY

AI Engineer with 5+ years of experience building production-grade AI systems spanning document digitization, retrieval-augmented generation (RAG), and machine learning applications. Specialized in designing LLM-powered workflows, vision-language extraction pipelines, and hybrid retrieval systems that operate reliably on noisy real-world enterprise data. Experienced in taking AI solutions from experimentation to production, including evaluation frameworks, on-prem deployment, model optimization, and compliance-sensitive infrastructure. Proven ability to replace manual and rule-based processes with scalable AI systems that improve operational efficiency and decision-making across logistics, knowledge management, and business operations.

## TECHNICAL SKILLS

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|-------------------------------------------|---------------------------------------------------------------|
| • <b>Programming &amp; APIs:</b>          | Python, SQL, FastAPI                                          |
| • <b>Models:</b>                          | VLM's (Qwen3-VL, PaddleOCR-VL), LLM's (Gemma, Mistral)        |
| • <b>Model Deployment:</b>                | vLLM, llama.cpp, Quantization                                 |
| • <b>OCR &amp; Document Processing:</b>   | PaddleOCR, PP-Structure, Tesseract, Layout Analysis           |
| • <b>AI Frameworks:</b>                   | Transformers, LangChain, RAG, LangGraph                       |
| • <b>Machine Learning:</b>                | Scikit-Learn, NLP, Sentiment Analysis, Feature Engineering    |
| • <b>Databases:</b>                       | PostgreSQL, ChromaDB, FAISS, Qdrant, MongoDB                  |
| • <b>Cloud &amp; Infrastructure:</b>      | Google Cloud Platform (Document AI, Vertex AI, Cloud Storage) |
| • <b>Evaluation:</b>                      | Eval Harness, Regression Testing, RAGAS                       |
| • <b>Infrastructure &amp; Deployment:</b> | Docker, RunPod, Streamlit                                     |
| • <b>Data Analysis:</b>                   | NumPy, Pandas, Tableau                                        |
| • <b>Claude:</b>                          | Skills, Sub-Agents, Hooks, MCP Servers, Plugins               |

## EXPERIENCE

PhFlow AI

Tampa, USA

AI Engineer

July 2025 – Present

- Replaced a template-based document extraction system with a custom LLM reasoning pipeline, cutting processing time from ~16s to under 10s while handling significantly more vendor layout variations.
- Designed schema-constrained prompting with explicit negative constraints to prevent field hallucination, reducing extraction errors on production shipment documents.
- Introduced a two-pass evaluation harness testing extraction across document variants and edge cases, catching regressions before production rollout and replacing manual spot-checking entirely.
- Integrated an AI assistant using RAG into warehouse operations, giving floor teams natural language access to shipment status, inventory levels, and operational knowledge without digging through source systems.
- Built a POC hybrid retrieval application using LangGraph, combining BM25 and vector retrieval to prototype expanded document search capabilities beyond the current warehouse assistant.
- Deployed inference on private GPU servers via Docker with no external API calls, keeping sensitive logistics documents on-prem to meet a hard client data governance requirement.
- Configured GitHub MCP server access for AI agents, automating PR reviews, issue triage, and repository navigation across active codebases.

GanaIT

Tampa, USA

AI Engineer

May 2023- June 2025

- Developed a two-stage query router combining regex rules and a fine-tuned classifier, hitting ~95% routing accuracy across multiple knowledge domains without using a pre-built agent framework.
- Owned the complete retrieval pipeline across 12,000+ pages data, using MongoDB to store document metadata and hierarchical structure alongside Qdrant's vector index for faster filtering and context reconstruction.
- Improved retrieval quality by combining BM25 keyword search, vector retrieval, and cross-encoder re-ranking, addressing failure cases on short and ambiguous queries where dense search alone fell short.
- Built an evaluation framework using RAGAS to score retrieval precision and answer relevance across varied query types, giving the team a clear, measurable way to track quality over time.
- Moved from hosted API inference to local llama.cpp with Qdrant and BGE embeddings running entirely on-prem, eliminating external data exposure for a client with strict data residency requirements.
- Built context-aware response generation on top of retrieved chunks, tuning prompt structure and context window usage to improve answer relevance across query types that varied widely in specificity.

- Built an NLP pipeline to process thousands of customer reviews and support comments, classifying sentiment and grouping recurring complaints into categories that product and ops teams could act on.
- Cleaned and prepared raw feedback text for model training, handling inconsistent formatting, noisy input, and ambiguous labels that were common across real customer-submitted data.
- Compared multiple classification algorithms on the same evaluation sets to select the best performing model, replacing subjective decisions with measurable benchmarks.
- Ran topic grouping across complaint categories to surface recurring issues, giving managers a summarized view of what customers were unhappy about without reading individual comments.
- Evaluated model predictions against business outcomes rather than just accuracy scores, validating that the pipeline was surfacing actionable signals rather than statistically clean but useless results.
- Delivered reports showing complaint category distribution and sentiment trends over time, enabling teams to track whether specific issues were growing or resolving across reporting periods.

- Consolidated reporting across multiple disconnected operational systems into a unified data layer using Pandas, eliminating multi-day manual reconciliation cycles.
- Automated recurring SQL and Python reporting workflows, reducing the time spent on manual report generation and freeing up time for analysis work.
- Traced data mismatches to root causes across source systems and put validation checks in place, catching issues that previously went undetected until they appeared in reports.
- Built placement and trainee progress dashboards using Tableau, replacing ad-hoc spreadsheet pulls with consistent live visibility for program managers, updated weekly.
- Ran cohort analysis across training programs to surface performance gaps in specific tracks that were invisible in aggregate numbers, with findings feeding directly into program restructuring decisions.
- Partnered with program teams to translate ambiguous reporting requests into defined metrics and analytical workflows, ensuring the output answered the business question behind the mask.

EDUCATION

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